ANGULAR MOMENTUM

Chapter

3

Objectives

- Define the angular momentum of a system rotating about a fixed axis.
- Apply the relation between the angular momentum and angular velocity.
- State the theorem of angular momentum.
- State the law of conservation of angular momentum.
- Explain some applications using the conservation of angular momentum.

By bringing her initially stretched arms back to her chest, a rotating skater can increase her rotational speed noticeably (Figure 3.1).
A horizontal disk turns around its vertical axis with a constant angular velocity. If we suddenly put a certain object on it, we notice that the angular velocity decreases. (Figure 3.2).
In order to explain this and other similar observation, scientists have defined a physical quantity, the angular momentum, that we will study in this chapter.

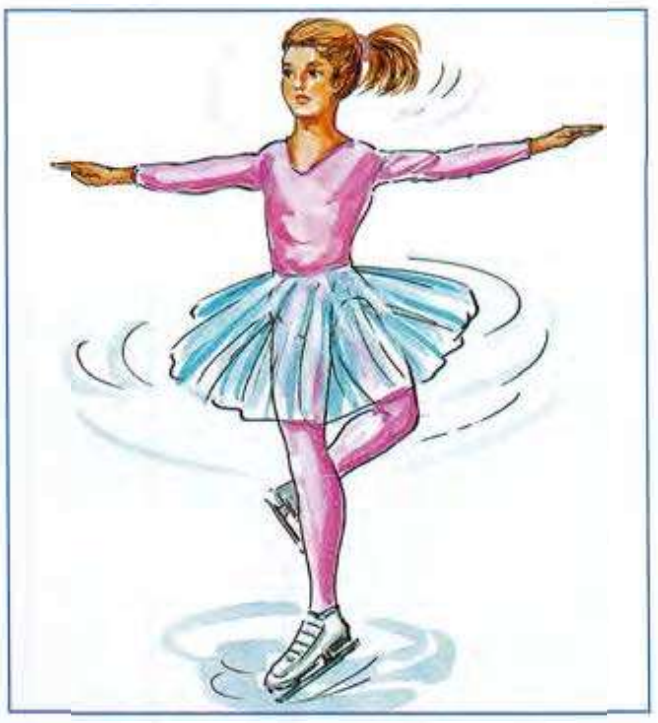

Figure 3.1 - A skater who spins on one point.

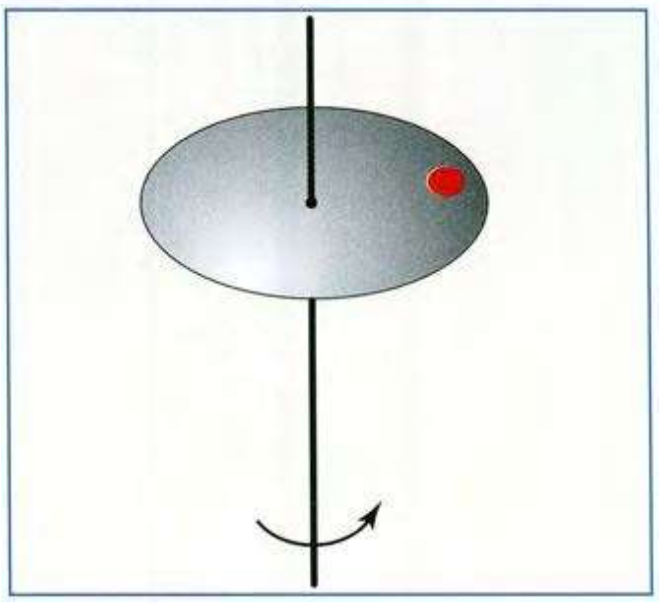

Figure 3.2

3.1 ANGULAR MOMENTUM OF A PARTICLE ROTATING ABOUT A FIXED AXIS

A particle A of mass m rotates about a fixed axis (Δ). We choose an arbitrary positive sense (Figure 3.3).

θ' is the angular velocity of the particle at a given instant, r its distance to the axis and $v = r\theta'$ the algebraic value of its linear velocity $\vec{v}$. Its linear momentum $\vec{p}$, at the given instant, is represented by the vector $m\vec{v}$ whose

- Line of action is tangent to the trajectory.
- Direction is that of the motion.
- Algebraic value is equal to $mv = mr\theta'$.

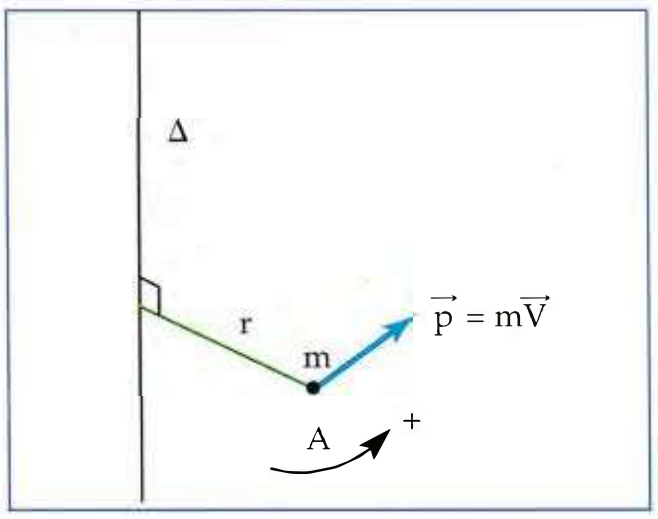

Figure 3.3

The algebraic value σ of the angular momentum of a particle with respect to the fixed rotational axis (Δ) is equal to the product of mv times the distance r separating the axis (Δ) from the line of action of $m\vec{v}$. $\boldsymbol{\sigma = mvr = mr\theta' r = mr^2\theta'}$

$$\Rightarrow \boldsymbol{\sigma = I\theta'}$$

where $I = mr^2$ is the expression for the **moment of inertia** of the particle of mass m with respect to (Δ).
In S.I units, I is expressed in $Kg.m^2$, θ' in rd/s and σ in $Kg.m^2/s$.

σ is positive if A rotates in the positive direction (Figure 3.4)

*According to the relation $S = r\alpha$, where S is expressed in m, r in m and α **respresents the numerical value of the angle.***
In fact: $2\pi r(m) \rightarrow 2\pi(rd)$
$S(m) \rightarrow \alpha(rd)$
$S = 2\pi r(m).\alpha(rd)/2\pi(rd)$
$\Rightarrow S(m) = r(m).\alpha$

and negative if A rotates in the negative direction (Figure 3.5).
From now on, we mean by angular momentum its absolute value.

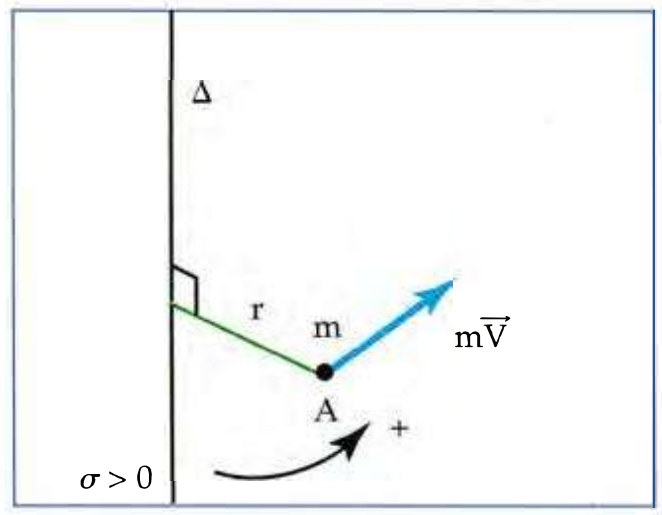

Figure 3.4

3.2 ANGULAR MOMENTUM OF A SYSTEM ROTATING ABOUT A FIXED AXIS

A system (S) is formed of many particles rotating about a fixed axis (Δ).

The angular momentum of the system (S) with respect to the axis (Δ), at any instant t, is equal to the sum of the angular momenta of all its particles with respect to (Δ).

$$\sigma = m_1 r_1^2 \theta'_1 + m_2 r_2^2 \theta'_2 + \ldots = \Sigma(m_i r_i^2 \theta'_i)$$

In the special case of a solid rotating about a fixed axis (Figure 3.6), all the particles of the solid have the same angular velocity θ' at any instant.

$$\sigma = \Sigma(m_i r_i^2 \theta') = \theta' \Sigma(m_i r_i^2) = I\theta'$$

where $I = \Sigma(m_i r_i^2)$ is the expression for the moment of inertia of the solid with respect to the axis (Δ).

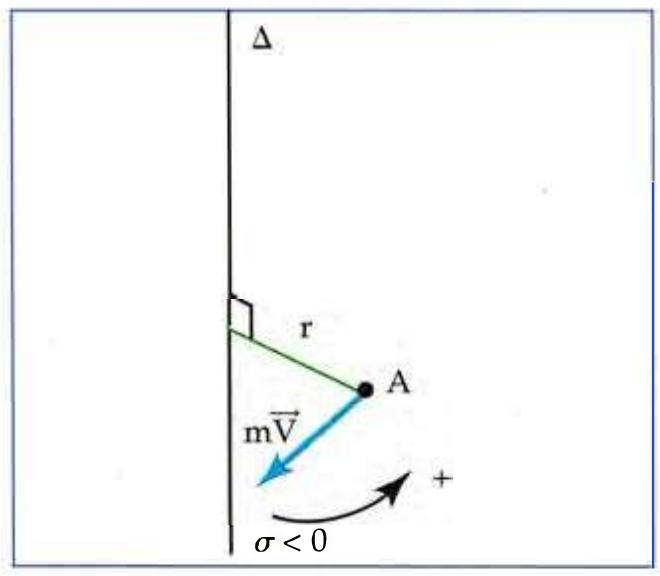

Figure 3.5

Application 3.1

The planet Neptune, of mass $m = 10^{26}$ Kg, describes around the sun an approximately circular trajectory of radius $r = 5 \times 10^9$ Km.
The period of revolution around the sun is T=165 years. Being very far from the Sun, Neptune is considered as a particle. Calculate the angular momentum σ of the planet with respect to an axis passing through the center of the sun and perpendicular to the plane of the trajectory.

Solution

The angular momentum of a particle with respect to a fixed axis is given by the formula: $\sigma = I\theta'$

where $I = mr^2 = 10^{26} \times (5 \times 10^{12})^2 = 25 \times 10^{50}$ Kg.m²

and $\theta' = \dfrac{2\pi}{T} = \dfrac{6.28}{165 \times 365 \times 24 \times 3600} = 328 \times 10^{-7}$ rd/s

$\Rightarrow \sigma = 82 \times 10^{43}$ Kg.m²/s

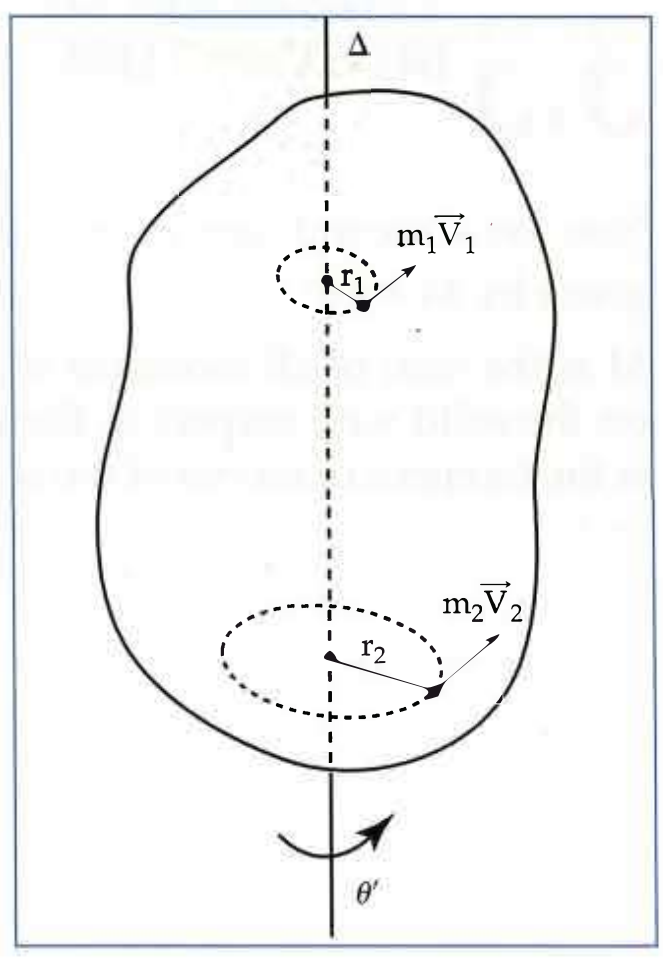

Figure 3.6

Application 3.2

Two particles rotate about the same fixed axis (Δ). They have with respect to (Δ) the same moment of inertia $I = 2 \times 10^{-3}$ Kg.m².
The first particle rotates with an angular velocity of 5rd/s in the positive direction, and the second rotates with an angular velocity of 8rd/s in the opposite direction.

a) Determine the angular momentum of each particle with respect to (Δ).

b) Determine the angular momentum of the system formed of the two particles with respect to (Δ).

Solution

a) $\sigma_1 = I_1\theta'_1 = 2 \times 10^{-3} \times 5 = 10^{-2}$ Kg.m²/s;

$\sigma_2 = I_2\theta'_2 = 2 \times 10^{-3} \times (-8) = -1.6 \times 10^{-2}$ Kg.m²/s.

b) The angular momentum of the system is equal to the sum of the angular momenta of the particles

$\sigma = \sigma_1 + \sigma_2 = 10^{-2} + (-1.6 \times 10^{-2})$

$= -0.6 \times 10^{-2}$ Kg.m²/s.

3.3 THEOREM OF ANGULAR MOMENTUM

Newton's second law of motion for solids in rotation is given by $M = I.\theta''$

M is the sum of all moments of the external forces applied on the solid with respect to the axis of rotation. I (constant) is the moment of inertia of the solid with respect to this axis, and θ'' is the angular acceleration of the solid.

The angular momentum of the solid with respect to the axis of rotation is $\sigma = I.\theta'$

Let us find the derivative with respect to time:

$$\frac{d\sigma}{dt} = I\frac{d\theta'}{dt} \Rightarrow \qquad \mathbf{M = \frac{d\sigma}{dt}}$$

This result is also valid in the case of a deformable system.

Theorem of angular momentum

The derivative, with respect to time, of the angular

momentum of a system of particles with respect to a fixed axis (Δ) is equal to the sum of the moments of the external forces applied on the system with respect to the same axis.

Application 3.3

A flywheel, formed of a disk of mass m = 20 kg and radius r = 10 cm, rotates about its fixed horizontal axis (Δ) with a constant speed of 3000 r. p. m. A braking couple of constant moment M = −0.25 m.N is applied to it.

Calculate the time needed to stop it.

Solution

- **System to be studied:** the flywheel.
- **Axis of rotation:** (Δ).
- **Positive direction of rotation:** that of the flywheel.
- **External forces and their moments:**

  a - The weight of the flywheel, whose moment with respect to (Δ) is $M_1 = 0$ (the weight intersects with the axis (Δ)).

  b - The normal reaction of the axis (Δ) whose moment with respect to (Δ) is $M_2 = 0$, since it intersects with (Δ).

  c - The braking couple whose moment with respect to (Δ) is M′.

- **Theorem of angular momentum:**

$$\frac{d\sigma}{dt} = M_1 + M_2 + M' = M' = -\ 0.25 \text{ m.N.}$$

Integrate with respect to time: $\sigma = -0.25\ t + \sigma_o$

- **Initial conditions:**

At $t_0 = 0$ (i.e when the brakes are applied)

The speed of rotation is $N_0 = 3000$ r.p.m = 50 r.p.s

The initial angular momentum of the flywheel is:

$$\sigma_0 = I\theta'_0 = \frac{mr^2}{2} \times 2\pi N_0 = 10\pi = 31.4 \text{ Kg.m}^2/\text{s}.$$

- **Value of the unknown quantity:**

$$\sigma = -\ 0.25\ t + 31.4$$

The flywheel stops when $\theta' = 0$, i.e when $\sigma = 0$ (Since $\sigma = I\theta'$).

$$\Rightarrow\ 0 = -\ 0.25\ t + 31.4\ \Rightarrow t = \frac{31.4}{0.25} = 125.6 \text{ s} \approx 126 \text{ s}$$

3.4 CONSERVATION OF ANGULAR MOMENTUM

If the sum of the moments of all external forces acting on a system is **zero** at any time t, then:

$$\sum_i \mathbf{M_i} = 0 \Rightarrow \frac{d\sigma}{dt} = 0 \Rightarrow \sigma = \text{constant at any time t.}$$

In such cases, **the angular momentum of the system is conserved.**

3.5 APPLICATIONS

a) A deformable system whose moment of inertia is variable:

A skater spins rapidly on one point as in figure 3.7. By folding her arms close to her breast the angular velocity increases substantially (Figure 3.8). Why does this thing happen?

Figure 3.7 - A ballet dancer spins on one toe with her arms outstretched.

Interpretation

The skater, considered as the mechanical system, rotates about a vertical axis (Δ). The external forces applied on the system (for, whether the arms are outstretched or folded to the body) are

a - the weight, whose moment with respect to (Δ) is zero.

b - the reaction of the ground on the skater, whose moment with respect to (Δ) is also zero. [The weight and the reaction are vertical forces and parallel to (Δ)]

- the frictional forces, considered negligible.

Therefore the angular momentum of the system is conserved.

$$\sigma_1 = \sigma_2 = \textbf{constant.}$$

Note that the muscular forces responsible for deforming the system, which are developed by the skater, are considered **internal forces**. They change the moment of inertia of the system about (Δ) from I_1 to I_2 and its angular velocity from θ'_1 to θ'_2.

Figure 3.8 - The same dancer spins on one toe with her arms down to her side.

First phase (arms outstretched):

σ_1 is the angular momentum of the skater with respect to

(Δ): $\sigma_1 = I_1.\theta'_1$.

Second phase (arms folded back on the breast):

σ_2 is the new angular momentum of the skater with respect to (Δ): $\sigma_2 = I_2.\theta'_2$.

The angular momentum is conserved:

$\Rightarrow \sigma_1 = \sigma_2 \quad I_1.\theta'_1 = I_2.\theta'_2$.

Since $I_2 < I_1$, therefore $\theta'_2 > \theta'_1$.

b) A deformable system whose moment of inertia is constant

Two horizontal rods (D_1) and (D_2) connected by means of a vertical torsion wire AB form one system. The system is suspended between two fixed points O and O′ by means of two vertical torsion-less strings OA and O′B (Figure 3.9).

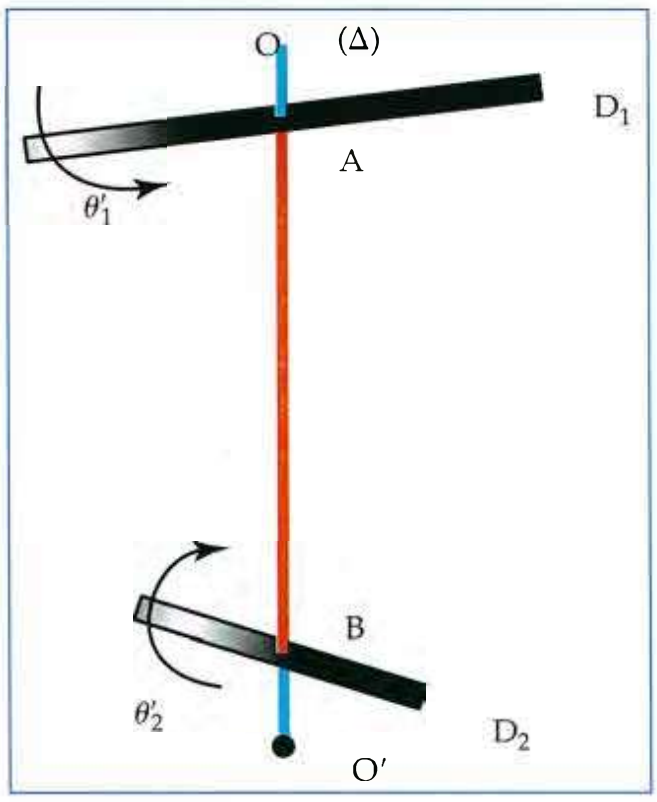

Figure 3.9

Rotate the two rods in opposite directions, each in a horizontal plane. The torsion wire AB is thus twisted. Release the rods without initial velocity. **They oscillate in opposite directions.**

Interpretation

The system rotating about the vertical axis OO′ is formed of the rods (D_1) and (D_2) and the torsion wire AB. At any instant, the external forces acting on the system are its weight and the tensions in the strings OA and O′B, whose moments with respect to the axis OO′ is zero (these forces are parallel to OO′).

At any instant, the angular momentum of the system is conserved. σ **= constant**

But the initial angular momentum of the system is $\sigma_0 = 0$ at $t_0 = 0$, since the system is released from rest.

Therefore, at any instant during oscillations $\sigma = 0$. I_1 and I_2 are respectively the moments of inertia of (D_1) and (D_2) with respect to OO′ (we ignore that of AB with respect to OO′).

At instant t, the angular velocities of (D_1) and (D_2) are θ'_1 and θ'_2 respectively. During oscillations

$$\sigma = 0 = \sigma_1 + \sigma_2 = I_1\theta'_1 + I_2\theta'_2$$

$$\Rightarrow \theta'_1 = -\frac{I_2}{I_1}\theta'_2$$

This relation shows that θ'_1 **and** θ'_2 **always have opposite signs, and the rod whose moment of inertia is small rotates faster than the other.**

3.6 COMPARISON TABLE OF TRANSLATIONAL AND ROTATIONAL MOTIONS

Translational motion	Rotational motion
Linear abscissa (elongation)	Angular abscissa
Linear velocity	Angular velocity
Linear acceleration	Angular acceleration
Mass (Inertia)	Moment of inertia
Force	Moment of a force
$\frac{d(\vec{p})}{dt} = \Sigma\vec{F}_{ext}$	$\frac{d(\sigma)}{dt} = \Sigma M_{ext}$
Linear momentum	Angular momentum
$\Sigma\vec{F}_{ext} = \vec{0} \Rightarrow \vec{p}$ is conserved	$\Sigma M_{ext} = 0 \Rightarrow \sigma$ is conserved.

SUMMARY

- The angular momentum of a particle with respect to an axis of rotation is $\sigma = I\theta'$.
- The angular momentum of a system of particles with respect to an axis of rotation is equal to the sum of the angular momenta of its particles with respect to this axis.
- In the case of a solid, the angular momentum is $\sigma = I\theta'$.
- The derivative, with respect to time, of the angular momentum of a system of particles with respect to a fixed axis (Δ) is equal to the sum of the moments of external forces applied on the system, with respect to (Δ).
- When the resultant moment of external forces acting on a system is zero, with respect to the axis of rotation, the angular momentum of the system is conserved.